

[illegible]

May 2025

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- 1-

10. **Multicasting is a method of sending data to:**
 A) A single destination B) Broadcast addresses
 C) Multiple destinations D) Single Address
11. **IPv6 addresses are represented as:**
 A) 64-bit decimal numbers B) 32-bit binary numbers
 C) 128-bit hexadecimal numbers D) 48-bit hexadecimal numbers
12. **What is the significance of subnetting in IPv4 addressing?**
 A) It increases the number of available IP addresses B) It simplifies network management
 C) It improves network security D) It simplifies data management
13. **BGP (Border Gateway Protocol) is an example of a:**
 A) Distance-vector routing protocol B) Link-state routing protocol
 C) Multicast routing protocol D) Path-vector routing protocol
14. **Distance-vector routing algorithms rely on routers sharing information about:**
 A) The entire network topology B) The cost to reach each destination from their directly connected neighbors
 C) The link states of all their interfaces D) The path vectors to all destinations
15. **In link-state routing, each router builds a:**
 A) Distance vector table B) Path vector table
 C) Link-state database D) Routing table based on neighbor information only
16. **Which routing protocol is commonly used on the Internet?**
 A) RIP B) BGP
 C) EIGRP D) OSPF
17. **Which of the following is NOT a standard application layer protocol?**
 A) HTTP B) FTP
 C) SMTP D) RIP
18. **Which protocol is used for transferring files between a client and a server?**
 A) FTP B) SMTP
 C) HTTP D) DNS
19. **What is the purpose of DNS?**
 A) To translate domain names into IP addresses B) To manage network resources
 C) To encrypt web traffic D) To provide authentication services
20. **Which protocol is used for sending email messages?**
 A) HTTP B) SMTP
 C) FTP D) DNS

PART - B: DESCRIPTIVE ANSWER QUESTIONS

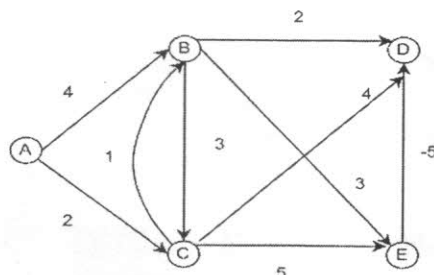
Unit – I		Marks	BT*	CO*	PO*
1.	a) Define data communication? Explain four fundamental characteristics of data communication.	8	L*2	1	1
	b) We need to send 265 kbps over a noiseless (ideal) channel with a bandwidth of 30 kHz. How many signal levels do we need?	4	L3	2	2
	c) Explain analog to analog conversion techniques in detail.	4	L2	2	1
2.	a) Define TCP/IP suite? Explain in detail layered architecture in the TCP/IP protocol suite.	8	L2	1	1
	b) A network using CSMA/CD has a bandwidth of 20 Mbps. If the maximum propagation time is 45.7 μ s, what is the minimum size of the frame?	4	L3	2	2
	c) What is Ethernet? Explain the standard Ethernet frame format.	4	L2	2	2

3. a) What are the three ways of communication between two devices and explain.
 b) Explain five components of a data communications system.
 c) What is CSMA? What should a channel do if channel is busy?

4	L2	1	1
4	L2	1	1
8	L2	2	2

Unit – II

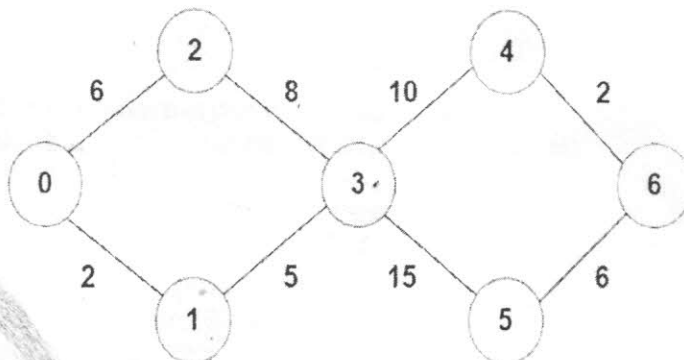
4. a) Explain Path vector algorithm with example.
 b) Define IPv4 fragmentation? Illustrate with example.
5. a) Apply Bellman Ford Algorithm for the below shown graph. Find out distance vectors from the node A and illustrate algorithm in detail.



- b) Explain IPv6 extension header types.

8	L3	4	2
8	L2	3	1

6. a) Explain in detail with example IPv4 classless addressing.
 b) Apply Dijkstra's algorithm for the below graph and illustrate algorithm in detail.



8	L3	4	2
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Unit – III

7. a) Explain HTTP request and response message formats with example.
 b) What are the services provided at the transport layer.
8. a) What is TCP segment? Explain TCP segment format in detail.
 b) Write a note on: i) Client-server paradigm ii) Peer-to-peer paradigm.

8	L2	5	2
8	L2	5	2
8	L2	5	2
8	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
